

PERSONAL GENOMICS REPORT

Minko

Genetic Health Narrative

43

TRAITS ANALYSED

17

ELEVATED SIGNALS

0

FAVOURABLE SIGNALS

July 17, 2026

GENERATED

Important: This report is for personal educational purposes only — not a medical diagnosis. Genetic predisposition does not determine health outcomes. Please discuss any findings of concern with a qualified healthcare professional.

Your Genetic Story — The Big Picture

Your Genetic Story — The Big Picture

After looking at your complete genetic profile, a coherent story emerges. You do not have any dramatic single-gene disorders, but you do have moderate genetic loading across several interconnected systems. The pattern is clear:

- **Cardiometabolic system** — moderate tendency toward higher lipids, blood pressure sensitivity, and metabolic efficiency challenges.
- **Immune & inflammatory regulation** — consistent elevation across multiple autoimmune and inflammatory conditions.

- **Brain & cognitive health** — signals in mood regulation, neurodegeneration risk, and cognitive performance under stress.

The common biological theme running through your profile is **inflammation and metabolic resilience**. Your body appears to run a bit "hot" in inflammatory pathways and is less efficient at certain protective and repair processes. This is not uncommon, and importantly, it is highly modifiable through lifestyle.

The highest-leverage areas for you are:

1. **Consistent physical activity** — especially aerobic + resistance training. This positively influences all three major systems (cardiometabolic, immune, and cognitive).
 2. **Anti-inflammatory nutrition** — Mediterranean-style eating, high omega-3 intake, adequate Vitamin D, and magnesium.
 3. **Sleep and stress management** — critical for immune regulation and cognitive health.
- Your genetics also give you clear advantages in other areas (height, intelligence, some protective variants). The overall picture is not one of inevitable disease, but of specific vulnerabilities that respond well to targeted lifestyle intervention.

This is your map. The terrain is known. What you do with it is up to you.

Physical Appearance & Traits

Eye Color

LIKELY BLUE-GREEN

GWAS Summary

Likely blue-green for Eye Color. The GWAS polygenic score is +0.0 (above population average), with 6 loci showing elevated risk allele dosage. Key genes implicated include OCA2, HERC2, SLC24A4.

Based on 6 of 6 IrisPlex marker SNPs present on your chip, there is moderate agreement toward blue. The single strongest marker, rs12913832 (GG), points blue, and this locus

alone drives most of IrisPlex's real-world predictive accuracy. Genotype and direction of effect for each marker is listed below.

Deeper Analysis

Your eye colour prediction sits at the blue-green boundary, which is exactly where your genotype combination places you. The dominant signal is rs12913832 GG — this single variant in the HERC2 gene is the strongest known predictor of eye colour in humans, and GG is the classic blue-associated genotype found in roughly 97% of blue-eyed Europeans. On its own, this would predict blue.

What introduces the green element is the combination of secondary signals pulling in different directions. SLC24A4 GT and IRF4 CT are both intermediate — they neither fully reinforce the blue signal nor push toward brown. SLC45A2 GG is the ancestral allele, which doesn't contribute the European lightening variant, adding a small counterweight. TYR AG (heterozygous) reduces melanin synthesis modestly across all tissues including the iris.

The net result: a strongly reduced OCA2-driven melanin signal from HERC2, partially modulated by intermediate alleles at supporting loci. This produces a lighter-than-brown iris with the specific hue sitting somewhere between blue and green depending on lighting conditions and iris structure. Blue-green or green-blue are both plausible phenotypic outcomes. Brown is effectively ruled out — you carry none of the classic brown-promoting allele combinations at the key loci.

One practical note: TYR CA's melanin-reducing effect extends beyond eye colour. The same enzyme reduction that lightens the iris also reduces UV-protective eumelanin in skin. This is not merely a cosmetic footnote — it has genuine implications for sun damage accumulation over a lifetime, addressed further in the skin section.

Hair Color

MEDIUM TO DARK BROWN

GWAS Summary

Medium to dark brown for Hair Color. The GWAS polygenic score is +0.0 (above population average), with 7 loci showing elevated risk allele dosage. Key genes implicated include MC1R Arg160Trp, MC1R Cys289Arg, MC1R Arg163Gln.

Medium to dark brown — No MC1R red hair variants detected — red hair is ruled out. SLC45A2 GG ancestral allele is the dominant pigmentation signal — dark hair baseline. TYR CA reduces melanin output modestly, consistent with the overall pigmentation profile. 7 SNPs combined.

Deeper Analysis

Your hair colour genetics tell a clean, consistent story: medium to dark brown, with no red component. The MC1R gene — which functions as the primary switch between red/yellow pheomelanin and dark eumelanin — shows no risk variants across all four tested positions (Arg151Cys, Arg160Trp, Arg163Gln, Cys289Arg). This effectively rules out red hair entirely. You inherited a fully functional MC1R receptor, which means your melanocytes default to eumelanin production.

The dominant pigmentation signal is SLC45A2 GG. This is the ancestral allele — common in non-European and darker-pigmented populations — which does not contribute the European hair-lightening variant. Combined with the absence of MC1R variants, this firmly establishes a dark baseline.

The only modest lightening signal comes from TYR AC (heterozygous at rs1042602), which reduces tyrosinase enzyme efficiency. This is a moderate effect — it may contribute some softening of the darkest possible hue, which is why the prediction is medium-to-dark rather than jet black. SLC24A4 GT is intermediate and adds no meaningful lightening.

The overall picture: dark brown hair driven by intact eumelanin machinery, with a modest TYR-mediated reduction that prevents the very darkest end of the spectrum. Environmental factors (sun exposure causing pheomelanin accumulation, ageing) will modify this over time, but the genetic baseline is clearly dark.

Skin Tone**MEDIUM (TYPE III)****GWAS Summary**

Medium (Type III) for Skin Tone. The GWAS polygenic score is +0.0 (above population average), with 6 loci showing elevated risk allele dosage. Key genes implicated include counteracts SLC24A5.

Medium (Type III). SLC24A5 AA is the dominant signal — this locus alone explains the majority of pigmentation difference between European and African populations and strongly predicts lighter skin. However, SLC45A2 GG (ancestral allele) partially counteracts this, suggesting a light-to-medium result rather than very fair. TYR CA reduces melanin output further, adding freckling tendency and UV sensitivity. Combined picture: skin that tans moderately but burns more readily than average. Moderate UV sensitivity — burns before tanning. 6 SNPs combined.

Deeper Analysis

Your skin pigmentation genetics reveal an interesting tension between two opposing forces, which produces a medium rather than extreme result in either direction.

The lightening signal is substantial: SLC24A5 AA is the derived European allele at the single most influential skin colour locus known. This variant spread through European and South Asian populations roughly 8,000 years ago, likely as an adaptation to low-UV environments requiring efficient vitamin D synthesis. Its effect size is large enough that it alone explains a significant fraction of the pigmentation difference between European and West African populations.

Counterbalancing this is SLC45A2 GG — the ancestral allele at a second major pigmentation gene. Unlike the CC European variant at SLC45A2, GG does not contribute additional lightening. This creates a genuine partial offset: SLC24A5 pushes strongly toward lighter skin; SLC45A2 holds back from adding further lightening, producing a light-to-medium rather than very fair result.

TYR CA (heterozygous) adds a third layer: reduced tyrosinase activity means less eumelanin output across all skin cells. This contributes to freckling tendency and, critically, to reduced UV protection. The skin's tanning response depends on tyrosinase upregulation in response to UV — with one reduced-function TYR allele, this response is blunted. The practical consequence: your skin burns more readily than it tans, and UV damage accumulates faster than in someone with full tyrosinase activity. SPF50 is not overcautious for this profile — it is the appropriate baseline, used consistently rather than occasionally.

The Fitzpatrick Type III prediction (medium skin, tans after initial burn) fits this genotype combination accurately.

Hair Loss

LOW GENETIC HAIR LOSS RISK

GWAS Summary

Low genetic hair loss risk for Hair Loss. The GWAS polygenic score is +0.0 (above population average), with 3 loci showing elevated risk allele dosage. Key genes implicated include AR/EDA2R, AR.

Low genetic hair loss risk. GWAS: 1 elevated loci, PGS +2.8. Male pattern baldness is ~80% heritable — one of the most genetically determined traits in the report. The X-linked AR/EDA2R locus is inherited from your mother. 3 SNPs combined.

Deeper Analysis

Your hair loss genetics tell a reassuring story. The X-linked AR/EDA2R locus — the single strongest signal for male pattern baldness, with ORs above 2.0 in published GWAS — shows rs2497938 CC, which is the non-risk genotype. This locus is inherited maternally (on the X chromosome), meaning the signal you carry from your mother's side does not confer the classic androgenetic alopecia predisposition at this dominant position.

The GWAS confirms this: only 1 elevated locus out of 15 genotyped, with a PGS of +2.8 — the lowest meaningful polygenic score in your entire profile. Male pattern baldness is

approximately 80% heritable, making this one of the most genetically determined traits in the report. The low genetic signal here is therefore genuinely informative rather than a data gap.

AR receptor sensitivity and DHT conversion rates have additional genetic determinants (5-alpha reductase, androgen receptor CAG repeat length) that were not captured on your chip, so a small residual uncertainty remains. But the primary signal is clear and favourable.

Height

STRONG ABOVE-AVERAGE HEIGHT SIGNAL

GWAS Summary

Strong above-average height signal for Height. The GWAS polygenic score is +0.0 (above population average), with 3 loci showing elevated risk allele dosage. Key genes implicated include HMGA2, GDF5.

Strong above-average height signal. GWAS PGS +401.1 from 29 elevated loci. Height is ~80% heritable in well-nourished populations. The GWAS polygenic score of +401 is the second highest in your profile, consistently pointing toward above-average stature. Realised height depends substantially on childhood nutrition and health. 3 curated SNPs combined.

Deeper Analysis

Height is one of the most polygenic traits in the human genome — thousands of variants each contributing fractions of a millimetre. Your polygenic score of +401 from 29 elevated loci is the second-highest quantitative score in your profile (after intelligence), and it consistently points above the population mean.

HMGA2 rs1042725 TT is the one well-characterised curated locus on your chip. HMGA2 (High Mobility Group AT-hook 2) is a transcription factor that regulates growth and development — TT at this position is the taller-associated genotype, contributing approximately 0.4 cm per allele in large-scale studies.

The GWAS elevated loci include hits across pathways governing bone growth, IGF-1 signalling, and skeletal development. A PGS of +401 in the height GWAS (which uses beta coefficients in cm units) translates to a meaningful cumulative signal toward above-average stature. Height is approximately 80% heritable in well-nourished populations, and your genetic signal is consistent with above-average genetic potential for height — though actual realised height depends substantially on childhood nutrition, sleep, and health.

Cardiovascular Health

Cholesterol / LDL

MODERATELY ELEVATED LDL TENDENCY

GWAS Summary

Moderately elevated LDL tendency for Cholesterol / LDL. The GWAS polygenic score is +0.0 (above population average), with 7 loci showing elevated risk allele dosage. Key genes implicated include SORT1/CELSR2, PCSK9, APOE.

Moderately elevated LDL tendency. APOE status: APOE status: rs429358 not on chip — partial APOE data only. rs7412 CC — ϵ 2 signal, protective for LDL (lowers risk). PCSK9 and SORT1 variants show modest LDL-raising tendency. LPA rs10455872 AA — non-risk genotype at this locus (G is the risk allele). Diet and exercise are likely sufficient; monitor annually. 7 SNPs combined.

Deeper Analysis

Your cholesterol genetics present a mixed but ultimately moderate picture. The most important finding is what is absent: you carry rs7412 CC, which is the APOE ϵ 2 signal — associated with lower LDL and reduced cardiovascular risk compared to the common ϵ 3/ ϵ 3 genotype. This is a genuinely favourable finding at the most clinically significant cholesterol locus.

The note of caution is that rs429358 — the second SNP needed to fully call APOE genotype — was not on your chip. This means we cannot rule out a mixed $\epsilon 2/\epsilon 4$ genotype, which would partially offset the $\epsilon 2$ protection. A clinical APOE genotype test (inexpensive, available through your GP) would resolve this definitively and is worth doing given its implications for both cardiovascular and Alzheimer risk.

Beyond APOE, the LDL-relevant variants show: rs629301 TT at SORT1/CELSR2 carries two copies of the LDL-raising allele — this locus modestly elevates LDL through effects on hepatic SORT1 expression and LDL receptor trafficking. rs11591147 GG at PCSK9 shows no protective loss-of-function allele, meaning you don't have the natural PCSK9 inhibition that some individuals carry (which can lower LDL by 30% without medication). LPA rs10455872 AA is non-risk — you do not carry the Lp(a)-elevating variant, which is a meaningful protective finding since elevated Lp(a) is an independent cardiovascular risk factor not captured by standard lipid panels.

Overall: the SORT1 variant creates a modest LDL-raising tendency that is likely counterbalanced somewhat by the APOE $\epsilon 2$ signal. Diet and exercise remain the first-line approach. Annual lipid panels are appropriate, and resolving the full APOE status would meaningfully sharpen this picture.

Coronary Artery Disease

ELEVATED CAD RISK

GWAS Summary

Elevated CAD risk for Coronary Artery Disease. The GWAS polygenic score is +0.0 (above population average), with 7 loci showing elevated risk allele dosage. Key genes implicated include CELSR2, CDKN2B.

Elevated CAD risk. GWAS: 25 elevated loci, PGS +111.0. The 9p21 locus (CDKN2B-AS1) is the strongest common CAD variant — it acts independently of LDL. Standard cardiovascular risk factors (blood pressure, smoking, diabetes, LDL) interact multiplicatively with genetic risk. Statins and lifestyle modification are effective regardless of genotype. 7 SNPs combined.

Deeper Analysis

Coronary artery disease is where your genetic profile shows its most clinically significant elevation. The 9p21 locus (CDKN2B-AS1) — the single most replicated common CAD variant in the literature — appears twice in your data: rs4977574 AG (one risk copy) and rs1333049 CG (one risk copy at the secondary variant). These two variants are additive and their effects are well-established across multiple large cohort studies.

What makes the 9p21 locus particularly important is that it acts independently of LDL cholesterol. Its mechanism involves regulation of cell proliferation in vascular smooth muscle and inflammatory pathways — meaning standard lipid testing will not capture this risk. Someone with normal cholesterol and 9p21 risk alleles still has meaningfully elevated CAD risk compared to population baseline.

Additional signals: rs646776 TT at CELSR2/SORT1 (the same locus relevant to LDL) carries two copies of the risk allele and contributes via the LDL pathway. rs1122608 GG at LDLR is the LDL receptor region variant. The LPL rs264 GG is the protective allele — a gain-of-function variant that raises lipoprotein lipase activity and improves triglyceride clearance. This partial offset from LPL is a genuine counterbalancing factor.

The GWAS shows 25 elevated loci and a PGS of +111, confirming meaningful polygenic loading across multiple CAD pathways. Given this profile, attention to the full cardiovascular risk picture is warranted: blood pressure monitoring, regular lipid panels (with awareness that 9p21 risk is not captured by LDL alone), smoking avoidance, and physical activity. High-sensitivity CRP (hsCRP) testing is worth discussing with your GP as an inflammatory risk marker relevant to the 9p21 pathway.

Triglycerides

MODERATE TRIGLYCERIDE TENDENCY

GWAS Summary

Moderate triglyceride tendency for Triglycerides. The GWAS polygenic score is +0.0 (above population average), with 6 loci showing elevated risk allele dosage. Key genes implicated include APOA5, APOC3, LPL.

Moderate triglyceride tendency. GWAS: 57 elevated loci, PGS +113.8. APOA5 and LPL are the dominant loci. Dietary carbohydrate and alcohol intake interact strongly with genetic triglyceride risk — omega-3 supplementation and low-carbohydrate diets are particularly effective in carriers of APOA5 risk alleles. 6 SNPs combined.

Deeper Analysis

Your triglyceride genetics show a pattern of moderate elevation with one important protective counterweight. The primary risk signal is APOA5 AG (rs662799 heterozygous) — APOA5 is the dominant triglyceride locus, and the G allele impairs APOA5's role in activating lipoprotein lipase, leading to slower triglyceride clearance. Heterozygous carriers typically show mild-to-moderate triglyceride elevation, with the effect amplified substantially by carbohydrate intake and alcohol.

APOC3 AG (rs2266788) adds a secondary signal — APOC3 normally inhibits LPL, and the promoter variant increases APOC3 expression, further reducing triglyceride clearance. MTNR1B CG (heterozygous) affects fasting triglycerides via melatonin receptor signalling — a more modest contribution.

The key protective finding is LPL rs328 CC — the S447X gain-of-function variant. This is genuinely protective: the C allele creates a truncated but hyperactive form of lipoprotein lipase that substantially improves triglyceride clearance. This is one of the few cases in lipid genetics where a variant provides real biological protection, and its presence meaningfully counterbalances the APOA5 and APOC3 risk signals.

Dietary implications are direct: your APOA5 heterozygosity means carbohydrate and alcohol intake have an amplified effect on your triglyceride levels compared to someone without this variant. Omega-3 supplementation (EPA/DHA, 2-4g daily) is particularly effective in APOA5 risk carriers — clinical trials show 20-30% triglyceride reduction. Low-carbohydrate diets are more effective for you than for the average person. The LPL protective allele means you have better baseline clearance capacity than your risk alleles alone would suggest.

Low genetic signal for Atrial Fibrillation. The GWAS polygenic score is +13.1 (above population average), with 1 loci showing elevated risk allele dosage. Key genes implicated include LINCo1875.

Low genetic signal for Atrial Fibrillation. GWAS polygenic score is +13.1 (above average); 1 loci show elevated risk allele dosage out of 141 genotyped. AF heritability ~20–30%; PITX2 locus is the strongest signal.

Stroke

MODERATE GENETIC SIGNAL

Moderate genetic signal for Stroke. The GWAS polygenic score is +31.4 (above population average), with 9 loci showing elevated risk allele dosage. Key genes implicated include TBX5, ADGRL3, CXCL8.

Moderate genetic signal for Stroke. GWAS polygenic score is +31.4 (above average); 9 loci show elevated risk allele dosage out of 206 genotyped. Stroke heritability ~40%; many variants overlap with CAD and AF. Blood pressure control is the most effective intervention.

Heart Failure

ELEVATED GENETIC SIGNAL

Elevated genetic signal for Heart Failure. The GWAS polygenic score is +61.4 (above population average), with 22 loci showing elevated risk allele dosage. Key genes implicated include CELSR2, LDLR, CAV1.

Elevated genetic signal for Heart Failure. GWAS polygenic score is +61.4 (above average); 22 loci show elevated risk allele dosage out of 49 genotyped. Heart failure is largely downstream of CAD, hypertension, and diabetes — primary prevention of those conditions is key.

Metabolic Health

MTHFR / B-Vitamin Metabolism

MODERATE FOLATE METABOLISM SIGNAL

GWAS Summary

Moderate folate metabolism signal for MTHFR / B-Vitamin Metabolism. The GWAS polygenic score is +0.0 (above population average), with 5 loci showing elevated risk allele dosage. Key genes implicated include MTHFD1, MTHFR.

Moderate folate metabolism signal. You carry both MTHFR C677T (GA heterozygous) and MTHFR A1298C (GT heterozygous) — compound heterozygosity that reduces MTHFR enzyme activity by approximately 50-60%. This impairs conversion of dietary folate to active 5-methylTHF and slows homocysteine clearance. Practical action: supplement with methylfolate (5-MTHF) rather than folic acid, and methylcobalamin rather than cyanocobalamin. Test homocysteine, folate, and B12 blood levels to confirm metabolic status. 5 SNPs combined.

Deeper Analysis

Your folate and B-vitamin metabolism genetics are the most clinically actionable nutritional finding in this report, and they deserve careful attention because they were missing from the main health report entirely.

You carry two MTHFR variants simultaneously — compound heterozygosity that is more significant than either variant alone:

MTHFR C677T (rs1801133) GA — heterozygous. The T allele reduces MTHFR enzyme activity by approximately 35% in heterozygotes. MTHFR (methylenetetrahydrofolate reductase) is the key enzyme that converts folate into the active form (5-methylTHF) used to recycle homocysteine back to methionine. Reduced activity means slower homocysteine clearance and potentially elevated plasma homocysteine — a cardiovascular and neurological risk factor.

MTHFR A1298C (rs1801131) GT — heterozygous. The C allele at this position reduces MTHFR activity through a different mechanism, affecting the regulatory domain of the enzyme. Alone, A1298C heterozygosity has modest effects. Combined with C677T heterozygosity (compound heterozygosity), the functional enzyme activity can be reduced by 50-60% — approaching the level seen in C677T homozygotes.

MTR A2756G (rs1805087) AA — this is the wild-type at the methionine synthase gene, meaning no additional impairment of the B12-dependent remethylation pathway from this locus.

MTRR A66G (rs1801394) GG — this is the AA wild-type at methionine synthase reductase (the enzyme that recycles the B12 cofactor for MTR). GG here means no MTRR-related impairment.

MTHFD1 (rs2236225) AG — heterozygous at the methylenetetrahydrofolate dehydrogenase gene, which feeds into the folate cycle. This adds a modest additional strain on the overall one-carbon metabolism pathway.

The practical implications of compound MTHFR heterozygosity are direct and well-supported by clinical evidence:

First, standard folic acid (the synthetic form in most supplements) requires conversion to active 5-methylTHF by MTHFR itself — a step that is inefficient in your case. The solution is to supplement with **methylfolate (5-MTHF)** directly, bypassing the impaired conversion step entirely. 400-800 mcg of methylfolate daily is the standard recommendation for MTHFR compound heterozygotes.

Second, B12 status matters more for you than for the average person. The homocysteine-to-methionine conversion requires both active folate and B12 as cofactors. Methylcobalamin (the active form of B12) rather than cyanocobalamin is preferable for the same bypass reasoning. Testing homocysteine, folate, and B12 blood levels is worth doing to establish your actual metabolic status — genetics tell you the predisposition, blood tests tell you whether it is expressing.

Third, the cardiovascular relevance: elevated homocysteine is an independent risk factor for coronary artery disease and stroke — two areas where your GWAS data already shows elevated signals. This is not coincidental; the MTHFR-homocysteine-cardiovascular connection is well-established and represents a direct intervention opportunity. Adequate methylfolate and methylcobalamin supplementation has been shown to normalise homocysteine in MTHFR carriers.

This is one area where a relatively simple nutritional intervention (switching supplement forms) has genuine mechanistic rationale specific to your genotype.

Type 2 Diabetes

ELEVATED T2D RISK

GWAS Summary

Elevated T2D risk for Type 2 Diabetes. The GWAS polygenic score is +0.0 (above population average), with 12 loci showing elevated risk allele dosage. Key genes implicated include KCNJ11, PPARG, TCF7L2.

Elevated T2D risk. GWAS polygenic score is +116.8 (above average across 26 elevated loci). Note: rs1631619 GG (OR 5.0) is a population-specific finding from a Korean cohort and may inflate the PGS for non-Korean ancestries. Genotype pattern: 4 loci homozygous risk, 7 heterozygous (one risk copy). Heterozygosity at TCF7L2, KCNJ11, FTO and others indicates moderate rather than severe genetic loading — you carry one risk copy at most major loci, not two. Important: T2D is 40–70% heritable but lifestyle factors (weight, diet, exercise) interact strongly — genetic risk is not destiny. 12 SNPs combined.

Deeper Analysis

Your Type 2 diabetes genetic profile deserves a careful reading because the headline number — elevated risk — obscures an important pattern in the detail.

You are heterozygous at almost every major T2D locus. TCF7L2 CT (one risk copy), KCNJ11 CT (one risk copy), SLC30A8 TC (one risk copy), CDKN2A/B CT (one risk copy), HHEX TC (one risk copy), IGF2BP2 GT (one risk copy), FTO CA (one risk copy). This consistent heterozygosity across seven independent loci is the defining feature of your profile. It means you have accumulated moderate genetic loading broadly, rather than severe loading at any single gene.

The three homozygous risk hits (rs76895963 TT, rs10741243 GG, rs1631619 GG) strengthen the signal, but rs1631619 GG deserves special mention: its OR of 5.0 comes from a Korean

cohort study. Population-specific effect sizes frequently do not replicate at full magnitude in other ancestries, and this variant should be interpreted with that caveat in mind.

TCF7L2 is the most important finding here. It is the strongest and most widely replicated common T2D variant, operating through the Wnt signalling pathway to impair beta-cell function. Heterozygous carriers have approximately 1.4x baseline risk. Crucially, TCF7L2 risk carriers respond particularly well to lifestyle intervention — the Diabetes Prevention Program trial showed that carriers who underwent lifestyle modification had the same T2D incidence reduction as non-carriers, meaning the genetic effect is not fixed.

The practical message: your genetic profile warrants attention to modifiable risk factors — weight, physical activity, dietary glycaemic load — but the predominantly heterozygous pattern means lifestyle intervention is genuinely effective at this level of loading. Fasting glucose and HbA1c monitoring every few years is appropriate regardless of symptoms.

BMI / Obesity Tendency

MODERATE GENETIC BMI TENDENCY

GWAS Summary

Moderate genetic BMI tendency for BMI / Obesity Tendency. The GWAS polygenic score is +0.0 (above population average), with 7 loci showing elevated risk allele dosage. Key genes implicated include FTO, NEGR1.

Moderate genetic BMI tendency. GWAS: 48 elevated loci, PGS +492.4. FTO and MC4R are the strongest common BMI loci. Importantly, BMI genetic effects are highly modifiable by physical activity — FTO risk allele carriers who exercise regularly show substantially attenuated obesity risk (Kilpeläinen et al. 2011). Genetic BMI tendency is not fixed destiny. 7 SNPs combined.

Deeper Analysis

Your BMI genetics show moderate loading, with the key finding being FTO heterozygosity. rs9939609 AT means you carry one copy of the FTO risk allele (A) — each A allele adds

approximately 0.4 kg/m² to average BMI, putting you at modest rather than strong genetic BMI elevation. The secondary FTO variant rs17817449 GT adds a small additional signal.

A critically important counterbalance: rs2815752 AG at NEGR1 carries the protective allele (A) in heterozygous form. NEGR1 is involved in neuronal growth and appetite regulation — the protective variant is associated with lower BMI, and its presence here partially offsets the FTO signal.

The most important thing to know about your FTO genotype is what the research shows about modifiability. The Kilpeläinen et al. 2011 meta-analysis (PLOS Medicine, 218,000 participants) demonstrated that FTO risk allele carriers who were physically active had BMI values essentially identical to non-carriers. The genetic effect of FTO on BMI was almost completely abolished by regular physical activity. This is one of the clearest gene-environment interaction findings in all of genomics — and it is directly actionable. The genetic tendency exists; whether it expresses depends substantially on activity level.

Gout

MODERATE GENETIC SIGNAL

Moderate genetic signal for Gout. The GWAS polygenic score is +24.2 (above population average), with 8 loci showing elevated risk allele dosage. Key genes implicated include ALDH2, MYL2, ABCG2.

Moderate genetic signal for Gout. GWAS polygenic score is +24.2 (above average); 8 loci show elevated risk allele dosage out of 195 genotyped. Gout heritability ~35–60%; SLC2A9 and ABCG2 are dominant uric acid loci. Dietary purine reduction and alcohol avoidance are effective interventions.

Chronic Kidney Disease

MODERATE GENETIC SIGNAL

Moderate genetic signal for Chronic Kidney Disease. The GWAS polygenic score is +18.6 (above population average), with 7 loci showing elevated risk allele dosage. Key genes implicated include CXCL8, ADGRL3, SLC13A3.

Moderate genetic signal for Chronic Kidney Disease. GWAS polygenic score is +18.6 (above average); 7 loci show elevated risk allele dosage out of 95 genotyped. CKD heritability ~45%; blood pressure and diabetes control are primary interventions.

Vitamin D Levels

MILD GENETIC SIGNAL

GWAS Summary

Mild genetic signal for Vitamin D Levels. The GWAS polygenic score is +11.4 (above population average), with 4 loci showing elevated risk allele dosage. Key genes implicated include GC, CYP2R1.

Mild genetic signal for Vitamin D Levels. GWAS polygenic score is +11.4 (above average); 4 loci show elevated risk allele dosage out of 71 genotyped. Vitamin D levels are ~50% heritable; GC and CYP2R1 are the major loci.

Deeper Analysis

Your Vitamin D genetics show a nuanced picture that goes beyond the "mild" polygenic label. While the overall GWAS score is only modestly elevated (+11.4), your specific variants reveal inefficiencies at two critical stages: transport and cellular uptake.

Transport stage (GC gene): You carry rs7041 CA (heterozygous) and rs4588 GG. This combination produces a mix of Vitamin D Binding Proteins, resulting in moderately reduced efficiency in delivering Vitamin D to your tissues. It is not severely impaired, but it lowers the amount of bioavailable Vitamin D compared to individuals with optimal GC genotypes.

Receptor stage (VDR gene): More significant are your VDR variants — rs731236 AA (TaqI) and rs7975232 CC (ApaI). These are well-studied markers associated with reduced Vitamin D receptor sensitivity. This means that even when Vitamin D reaches your cells, the receptors that should trigger calcium absorption and bone mineralization respond less effectively.

The combined effect creates a two-stage "volume reduction" on Vitamin D activity. You are not unable to process Vitamin D, but you have a higher physiological requirement to achieve the same level of calcium absorption and bone mineralization as someone with more efficient genetics. This aligns with your earlier observation about reduced calcium storage capacity in bones.

Practical implications: Aim to maintain higher serum Vitamin D levels (ideally 50–70 ng/mL). Because this is partly a receptor sensitivity issue, optimizing cofactors is crucial:

- **Magnesium** (300–400 mg/day) — required for Vitamin D activation.
- **Vitamin K2 (MK-7)** (100–200 mcg/day) — directs calcium into bones rather than soft tissues.

Regular blood testing (25(OH)D, calcium, PTH) is recommended to fine-tune your levels.

Brain, Cognition & Neurological Health

Cognitive Profile

WARRIOR COGNITIVE PROFILE (COMT VAL/VAL)

GWAS Summary

Warrior cognitive profile (COMT Val/Val) for Cognitive Profile. The GWAS polygenic score is +0.0 (above population average), with 2 loci showing elevated risk allele dosage. Key genes implicated include COMT, BDNF.

Warrior cognitive profile (COMT Val/Val). GWAS PGS +1064.5 from 8 elevated loci. COMT GG (Val/Val) — Warrior profile: higher dopamine breakdown in PFC, better executive function under load and stress. BDNF CT heterozygous — partial reduction in activity-dependent BDNF; sleep and aerobic exercise are particularly important for memory consolidation. Intelligence GWAS PGS is the highest in your profile at +1064, though this includes educational attainment and Alzheimer pleiotropic signals. 2 curated SNPs combined.

Deeper Analysis

Your cognitive genetics contain the highest polygenic score in this entire report: PGS +1064 from 8 elevated loci in the intelligence GWAS. Before interpreting this, the important methodological context: intelligence GWAS studies frequently include educational attainment as a proxy phenotype, and many hits reflect pleiotropic variants that affect both educational achievement and Alzheimer's risk via shared biological pathways. The PGS score therefore reflects a mixture of cognitive ability, educational propensity, and neurological resilience signals — not a pure intelligence measure.

The curated SNPs provide the most interpretable layer. COMT rs4680 GG is the Val/Val (Warrior) genotype — higher COMT enzymatic activity means faster dopamine breakdown in the prefrontal cortex. This produces better working memory performance and executive function under high cognitive load and stress, at the cost of slightly lower baseline dopamine tone and potentially reduced exploratory behaviour. Critically, Val/Val carriers show superior cognitive performance specifically under pressure and in complex task-switching — the profile associated with performance in demanding professional environments.

BDNF rs6265 CT (Val66Met heterozygous) is the one nuancing factor. The Met allele reduces activity-dependent BDNF secretion, which affects synaptic plasticity and learning consolidation. However, heterozygosity means one functional Val allele remains — this is a partial rather than full reduction. In practice, Val66Met heterozygotes show modestly reduced episodic memory performance in some studies but maintain normal working memory.

The combination — COMT Warrior profile with heterozygous BDNF — suggests a cognitive profile that excels under cognitive demand and structured tasks, while being somewhat more dependent than average on sleep quality and aerobic exercise for optimal memory consolidation. BDNF is one of the most exercise-responsive genes known: regular aerobic exercise upregulates BDNF expression and substantially compensates for the Met allele's reduced baseline secretion.

GWAS Summary

Average Alzheimer risk for Alzheimer's Disease. The GWAS polygenic score is +0.0 (above population average), with 6 loci showing elevated risk allele dosage. Key genes implicated include CLU.

Average Alzheimer risk. APOE status: $\epsilon 2/\epsilon 2$ or $\epsilon 2/\epsilon 3$ — very low Alzheimer risk (rs429358 not on chip). GWAS: 50 elevated loci, PGS +171.6. APOE genotype is by far the strongest genetic factor — it alone accounts for ~50% of genetic Alzheimer risk. Other loci (CLU, BIN1, CR1, PICALM) each contribute modestly. Lifestyle factors (cardiovascular health, cognitive engagement, sleep) meaningfully modify genetic risk. 6 SNPs combined.

Deeper Analysis

Your Alzheimer's risk profile contains arguably the most reassuring finding in this entire report. rs7412 CC corresponds to the APOE $\epsilon 2$ allele — and APOE $\epsilon 2$ is the strongest known protective variant against late-onset Alzheimer's disease. $\epsilon 2$ carriers have approximately 50% lower Alzheimer's risk compared to the common $\epsilon 3/\epsilon 3$ genotype, and are significantly overrepresented among cognitively healthy individuals aged 85 and older.

The caveat, as noted in the cholesterol section, is that rs429358 was not on your chip — so we cannot fully exclude an $\epsilon 2/\epsilon 4$ mixed genotype. However, even $\epsilon 2/\epsilon 4$ carriers have attenuated risk compared to $\epsilon 4/\epsilon 4$ homozygotes, and the protective signal from $\epsilon 2$ is real and substantial regardless.

The GWAS data shows 50 elevated loci and a PGS of +171.6, which appears to contradict the APOE finding. This apparent contradiction is important to understand: the GWAS TSV for Alzheimer's contains many study types including education-Alzheimer pleiotropic studies (variants that affect both educational attainment and Alzheimer risk via shared pathways). The APOE genotype, which is the dominant factor in clinical Alzheimer risk, is not always the dominant driver of GWAS polygenic scores in research studies. Weight APOE status heavily in your interpretation.

Secondary loci: CLU CC (rs9331888) is a modest risk signal at the clusterin gene, which is involved in amyloid clearance. This is a small effect relative to APOE. The other curated loci

(BIN1, CR1, PICALM) were either not on chip or showed no risk alleles in your data.

The overall Alzheimer picture is genuinely favourable given the $\epsilon 2$ signal. The most evidence-based protective behaviours — cardiovascular health maintenance, regular aerobic exercise, quality sleep (critical for amyloid clearance via the glymphatic system), and cognitive engagement — remain worthwhile regardless of genotype, and are especially effective in $\epsilon 2$ carriers who already have a protective biological baseline.

Depression

ELEVATED DEPRESSION GENETIC SIGNAL

GWAS Summary

Elevated depression genetic signal for Depression. The GWAS polygenic score is +0.0 (above population average), with 6 loci showing elevated risk allele dosage. Key genes implicated include TMEM161B, BDNF, NEGR1.

Elevated depression genetic signal. GWAS: 32 elevated loci, PGS +71.8. Depression heritability is ~35–40%; most genetic variance is polygenic with very small individual effects. BDNF Val66Met and COMT Val158Met are the best-studied candidates — but effect sizes are modest and gene-environment interaction (stress, trauma, social support) dominates. Genetic risk for depression is not predictive at the individual level. 6 SNPs combined.

Deeper Analysis

Your depression genetics tell a nuanced story that requires careful context to interpret correctly. The headline — elevated genetic signal with 32 GWAS loci and PGS +71.8 — reflects genuine polygenic loading. But depression genetics have important interpretive limitations that are as important as the numbers themselves.

The most personally relevant findings are in the curated SNPs. BDNF rs6265 CT means you carry one copy of the Val66Met variant — the T (Met) allele is associated with reduced activity-dependent BDNF secretion. BDNF is the brain's primary neurotrophic factor,

supporting neuronal survival, plasticity, and stress recovery. The Met allele has been associated with reduced hippocampal volume, increased anxiety response, and greater vulnerability to stress-induced mood changes. Importantly, BDNF levels are highly responsive to aerobic exercise — physical activity is one of the most robust BDNF upregulators known, and this is one mechanism by which exercise reduces depression risk.

COMT rs4680 GG is the Warrior profile (Val/Val) — higher dopamine breakdown in the prefrontal cortex, which typically means better stress resilience and executive function under pressure, at the cost of slightly lower baseline dopamine tone. For depression, this is generally neutral-to-protective rather than a risk factor.

NEGR1 AG (rs1545843) and TMEM161B TT (rs10514299) are the first wave of genome-wide significant depression hits, but their individual effect sizes are very small — important at the population level, less meaningful individually.

The overarching message: the BDNF Val66Met heterozygosity is the most interpretively meaningful finding here. It suggests somewhat higher biological sensitivity to stress and environmental adversity — not a predetermined outcome, but a profile that responds particularly well to protective factors: regular aerobic exercise (the most evidence-based BDNF upregulator), quality sleep, and strong social connection. These are not generic lifestyle advice — for your specific genotype, they have direct neurobiological relevance.

Adhd MILD GENETIC SIGNAL

Mild genetic signal for Adhd. The GWAS polygenic score is +10.4 (above population average), with 2 loci showing elevated risk allele dosage. Key genes implicated include MSR1, MIR3924.

Mild genetic signal for Adhd. GWAS polygenic score is +10.4 (above average); 2 loci show elevated risk allele dosage out of 141 genotyped. ADHD heritability ~70–80%; highly polygenic.

Autism MILD GENETIC SIGNAL

Mild genetic signal for Autism. The GWAS polygenic score is +8.2 (above population average), with 4 loci showing elevated risk allele dosage. Key genes implicated include QPCT, PKIA,

MYO9B.

Mild genetic signal for Autism. GWAS polygenic score is +8.2 (above average); 4 loci show elevated risk allele dosage out of 42 genotyped. Autism heritability ~80%; de novo variants play a major role.

Bipolar Disorder MODERATE GENETIC SIGNAL

Moderate genetic signal for Bipolar Disorder. The GWAS polygenic score is +16.7 (above population average), with 7 loci showing elevated risk allele dosage. Key genes implicated include PALB2, PARD6B, GFRA2.

Moderate genetic signal for Bipolar Disorder. GWAS polygenic score is +16.7 (above average); 7 loci show elevated risk allele dosage out of 190 genotyped. Bipolar disorder heritability ~70–80%; shares genetic overlap with schizophrenia.

Schizophrenia ELEVATED GENETIC SIGNAL

Elevated genetic signal for Schizophrenia. The GWAS polygenic score is +125.2 (above population average), with 29 loci showing elevated risk allele dosage. Key genes implicated include SLC1A1, MIR100HG, PCDH7.

Elevated genetic signal for Schizophrenia. GWAS polygenic score is +125.2 (above average); 29 loci show elevated risk allele dosage out of 965 genotyped. Schizophrenia heritability ~80%; major MHC/HLA component.

Parkinson MODERATE GENETIC SIGNAL

Moderate genetic signal for Parkinson. The GWAS polygenic score is +27.9 (above population average), with 9 loci showing elevated risk allele dosage. Key genes implicated include RNA5SP74, LINC00648, P2RX5.

Moderate genetic signal for Parkinson. GWAS polygenic score is +27.9 (above average); 9 loci show elevated risk allele dosage out of 180 genotyped. Parkinson's heritability ~27%; LRRK2 and SNCA are major monogenic loci.

Sleep Duration

MILD GENETIC SIGNAL

Mild genetic signal for Sleep Duration. The GWAS polygenic score is +7.8 (above population average), with 4 loci showing elevated risk allele dosage. Key genes implicated include C2CD3, KCNK9, MIR548A1HG.

Mild genetic signal for Sleep Duration. GWAS polygenic score is +7.8 (above average); 4 loci show elevated risk allele dosage out of 136 genotyped. Sleep duration heritability ~30–40%; circadian gene variants play a role.

Cancer Risk

Breast Cancer

ELEVATED GENETIC SIGNAL

Elevated genetic signal for Breast Cancer. The GWAS polygenic score is +80.9 (above population average), with 18 loci showing elevated risk allele dosage. Key genes implicated include MPHOSPH6, LINC00877, LINC01924.

Elevated genetic signal for Breast Cancer. GWAS polygenic score is +80.9 (above average); 18 loci show elevated risk allele dosage out of 806 genotyped. BRCA1/2 are high-penetrance but rare; common variants each contribute modestly.

Prostate Cancer

ELEVATED GENETIC SIGNAL

Elevated genetic signal for Prostate Cancer. The GWAS polygenic score is +61.2 (above population average), with 12 loci showing elevated risk allele dosage. Key genes implicated include KLK3.

Elevated genetic signal for Prostate Cancer. GWAS polygenic score is +61.2 (above average); 12 loci show elevated risk allele dosage out of 722 genotyped. Prostate cancer is ~57% heritable; 8q24 locus is the strongest common signal.

Colorectal Cancer

ELEVATED GENETIC SIGNAL

Elevated genetic signal for Colorectal Cancer. The GWAS polygenic score is +55.1 (above population average), with 10 loci showing elevated risk allele dosage. Key genes implicated include PTPA, MGAT4A, RASGRF2.

Elevated genetic signal for Colorectal Cancer. GWAS polygenic score is +55.1 (above average); 10 loci show elevated risk allele dosage out of 518 genotyped. CRC heritability ~35%; lifestyle (fibre, red meat, exercise) modifies genetic risk.

Melanoma

MODERATE GENETIC SIGNAL

Moderate genetic signal for Melanoma. The GWAS polygenic score is +14.1 (above population average), with 6 loci showing elevated risk allele dosage. Key genes implicated include SLC45A2.

Moderate genetic signal for Melanoma. GWAS polygenic score is +14.1 (above average); 6 loci show elevated risk allele dosage out of 162 genotyped. Melanoma heritability ~50%; MC1R variants and UV exposure interact strongly. Skin protection is highly effective.

Lung Cancer

ELEVATED GENETIC SIGNAL

Elevated genetic signal for Lung Cancer. The GWAS polygenic score is +31.6 (above population average), with 13 loci showing elevated risk allele dosage. Key genes implicated include CHEK2, CAMK2D.

Elevated genetic signal for Lung Cancer. GWAS polygenic score is +31.6 (above average); 13 loci show elevated risk allele dosage out of 208 genotyped. Lung cancer genetic risk is dominated by smoking — genetic variants modify but do not replace smoking as the primary risk factor.

Bladder Cancer

MILD GENETIC SIGNAL

Mild genetic signal for Bladder Cancer. The GWAS polygenic score is +13.0 (above population average), with 3 loci showing elevated risk allele dosage. Key genes implicated include KIAA0825, CLK3, UGT1A8.

Mild genetic signal for Bladder Cancer. GWAS polygenic score is +13.0 (above average); 3 loci show elevated risk allele dosage out of 73 genotyped. Bladder cancer ~50% heritable for common

variants; smoking is the strongest risk factor.

Immune & Inflammatory Conditions

Asthma

ELEVATED GENETIC SIGNAL

Elevated genetic signal for Asthma. The GWAS polygenic score is +40.4 (above population average), with 14 loci showing elevated risk allele dosage. Key genes implicated include TACR1, NPBWR1, ACOT7.

Elevated genetic signal for Asthma. GWAS polygenic score is +40.4 (above average); 14 loci show elevated risk allele dosage out of 588 genotyped. Asthma heritability ~60–80%; strong gene-environment (allergen) interaction.

Lupus

ELEVATED GENETIC SIGNAL

Elevated genetic signal for Lupus. The GWAS polygenic score is +49.4 (above population average), with 32 loci showing elevated risk allele dosage. Key genes implicated include STAT4.

Elevated genetic signal for Lupus. GWAS polygenic score is +49.4 (above average); 32 loci show elevated risk allele dosage out of 374 genotyped. Lupus heritability ~44%; strong HLA component.

Rheumatoid Arthritis

ELEVATED GENETIC SIGNAL

Elevated genetic signal for Rheumatoid Arthritis. The GWAS polygenic score is +40.4 (above population average), with 10 loci showing elevated risk allele dosage. Key genes implicated include HLA, SERTM1, DPP4.

Elevated genetic signal for Rheumatoid Arthritis. GWAS polygenic score is +40.4 (above average); 10 loci show elevated risk allele dosage out of 757 genotyped. RA heritability ~60%; HLA-DRB1 accounts for ~30% of genetic variance.

Inflammatory Bowel

ELEVATED GENETIC SIGNAL

Elevated genetic signal for Inflammatory Bowel. The GWAS polygenic score is +56.6 (above population average), with 18 loci showing elevated risk allele dosage. Key genes implicated include NUDT15, IL23R, HLA.

Elevated genetic signal for Inflammatory Bowel. GWAS polygenic score is +56.6 (above average); 18 loci show elevated risk allele dosage out of 851 genotyped. IBD heritability ~75% (Crohn's) / ~70% (UC); NOD2 is the strongest common risk gene.

Psoriasis

ELEVATED GENETIC SIGNAL

Elevated genetic signal for Psoriasis. The GWAS polygenic score is +32.0 (above population average), with 13 loci showing elevated risk allele dosage. Key genes implicated include TYK2, IL12B.

Elevated genetic signal for Psoriasis. GWAS polygenic score is +32.0 (above average); 13 loci show elevated risk allele dosage out of 310 genotyped. Psoriasis heritability ~60–70%; HLA-C*06:02 is the major risk allele.

Crohn Disease

ELEVATED GENETIC SIGNAL

Elevated genetic signal for Crohn Disease. The GWAS polygenic score is +48.6 (above population average), with 13 loci showing elevated risk allele dosage. Key genes implicated include TNFSF15, IL23R.

Elevated genetic signal for Crohn Disease. GWAS polygenic score is +48.6 (above average); 13 loci show elevated risk allele dosage out of 584 genotyped. Crohn's heritability ~75%; NOD2 and ATG16L1 are key loci.

Multiple Sclerosis

MODERATE GENETIC SIGNAL

Moderate genetic signal for Multiple Sclerosis. The GWAS polygenic score is +28.1 (above population average), with 5 loci showing elevated risk allele dosage. Key genes implicated include PADI1, HLA, TSBP1.

Moderate genetic signal for Multiple Sclerosis. GWAS polygenic score is +28.1 (above average); 5 loci show elevated risk allele dosage out of 347 genotyped. MS heritability ~50%; HLA-DRB1*15:01 is the strongest risk allele.

Longevity & Ageing

Longevity

MODERATE LONGEVITY SIGNALS

GWAS Summary

Moderate longevity signals for Longevity. The GWAS polygenic score is +0.0 (above population average), with 3 loci showing elevated risk allele dosage. Key genes implicated include CETP.

Moderate longevity signals. GWAS: 18 longevity loci, PGS +62.4. APOE and FOXO3 are the two most replicated longevity loci. FOXO3 G allele (if present) is associated with exceptional longevity across Japanese, American, Danish and other cohorts — it's the most robust non-APOE longevity signal known. CETP variants that raise HDL are also overrepresented in centenarians. Lifestyle (exercise, diet, sleep, not smoking) remains the dominant modifiable longevity factor. 3 SNPs combined.

Deeper Analysis

Your longevity genetics show moderate positive signals, with the APOE picture being the centrepiece. As noted in the Alzheimer section, rs7412 CC corresponds to the APOE ε2 allele — and ε2 is consistently overrepresented among centenarians in multiple independent cohort studies. The mechanism is multifactorial: lower LDL, reduced neuroinflammation, more efficient lipid metabolism, and lower Alzheimer risk all contribute to the survival advantage.

The CETP variants add genuine additional longevity signal. CETP rs3764814 CT (one protective C allele) and CETP rs5882 AG (one protective allele) both point toward higher

HDL cholesterol — and elevated HDL has been consistently associated with longevity in centenarian studies, particularly in Ashkenazi Jewish and Japanese cohorts. The CETP variants work by reducing cholesterol ester transfer activity, allowing HDL particles to remain larger and more functional for longer.

FOXO3 was not on your chip — this is the single most replicated longevity locus outside APOE, active across five independent national cohorts. It is worth checking via a clinical or research genetic test if longevity genetics are of particular interest to you.

The GWAS shows 18 longevity-associated loci with PGS +62.4. Combined with the APOE $\epsilon 2$ signal and CETP variants, the overall longevity picture is moderately favourable at the genetic level. The evidence across all longevity studies consistently shows that lifestyle factors — particularly avoiding smoking, regular physical activity, Mediterranean-style diet, and maintaining social engagement — explain more variance in reaching exceptional age than genetics does. Your genetics provide a reasonable starting position; what you do with it matters more.

Lifestyle, Nutrition & Drug Metabolism

Caffeine Metabolism

MILD GENETIC SIGNAL

Mild genetic signal for Caffeine Metabolism. The GWAS polygenic score is +5.5 (above population average), with 3 loci showing elevated risk allele dosage. Key genes implicated include CYP2A6.

Mild genetic signal for Caffeine Metabolism. GWAS polygenic score is +5.5 (above average); 3 loci show elevated risk allele dosage out of 33 genotyped. Caffeine metabolism is ~50% heritable; CYP1A2 is the primary enzyme gene.

Alcohol Consumption

ELEVATED GENETIC SIGNAL

Elevated genetic signal for Alcohol Consumption. The GWAS polygenic score is +37.8 (above population average), with 10 loci showing elevated risk allele dosage. Key genes implicated include

ADH1B, SLC39A8.

Elevated genetic signal for Alcohol Consumption. GWAS polygenic score is +37.8 (above average); 10 loci show elevated risk allele dosage out of 78 genotyped. Alcohol metabolism heritability ~50%; ADH1B and ALDH2 are the dominant variants. ALDH2 variants (common in East Asians) cause alcohol flush reaction.

Pharmacogenomics

MILD GENETIC SIGNAL

GWAS Summary

Mild genetic signal for Pharmacogenomics. The GWAS polygenic score is +6.4 (above population average), with 3 loci showing elevated risk allele dosage. Key genes implicated include JAK2, ZNF646, CPVL.

Mild genetic signal for Pharmacogenomics. GWAS polygenic score is +6.4 (above average); 3 loci show elevated risk allele dosage out of 32 genotyped. Drug metabolism varies substantially by genotype; CYP2D6, CYP2C19, and TPMT are clinically actionable.

Deeper Analysis

Your pharmacogenomics profile (from PharmGKB) shows potential interactions with several medications. Here are the most relevant ones based on your SNPs:

- **tacrolimus** — 452 variant matches
- **warfarin** — 375 variant matches
- **methotrexate** — 330 variant matches
- **methadone** — 181 variant matches
- **efavirenz** — 174 variant matches
- **imatinib** — 107 variant matches
- **methyphenidate** — 100 variant matches
- **risperidone** — 96 variant matches

Recommendation: Discuss these with your doctor or pharmacist before starting new medications, especially antidepressants, painkillers, or blood thinners.

Pharmacogenomic Insights

The following markers indicate potential variability in drug response based on your genotype.

Efficacy

Drug	Gene	Genotype
citalopram, escitalopram, fluoxetine, fluvoxamine, paroxetine, sertraline	FAIM2	TC
sitagliptin	KCNQ1	GT
sitagliptin	CDKAL1	GG
sitagliptin	CYP2C9	CC
exenatide	TCF7L2	TC
adalimumab	SLCO1C1	AA
peginterferon alfa-2a, peginterferon alfa-2b, ribavirin	IFNL3	TT
peginterferon alfa-2b, ribavirin	IFNL3	TT
corticosteroids	GLCCI1	CC
corticosteroids	GLCCI1	AA
corticosteroids	GLCCI1	GG

antiepileptics	SCN1A	TT
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Dosage

Drug	Gene	Genotype
heroin	OPRK1	TC
ethanol	FTO	TG
acenocoumarol	VKORC1	GA
remifentanil	ADRA2A	CC
remifentanil	OPRM1	GA
mercaptopurine, methotrexate	ABCC4	CC
warfarin	CYP4F2	CC
warfarin	CYP4F2	CC
warfarin	VKORC1	AC
methadone	ABCB1	GG
methadone	ABCB1	GG
methadone	CYP2B6	GT

Metabolism/PK

Drug	Gene	Genotype
tacrolimus	CYP3A4	CC

tacrolimus	ABCB1	CC
tacrolimus	ABCB1	GG
simvastatin acid	SLCO1B1	CC
simvastatin acid	SLCO1B1	GA
rifampin	SLCO1B1	CC
isoniazid	nan	AG
pyrazinamide	AOX1	AA
voriconazole	ABCB1	GG
posaconazole, tacrolimus	ABCB1	GG
rivaroxaban	ABCB1	TT
rivaroxaban	ABCB1	GG

nan

Drug	Gene	Genotype
letrozole	CYP2A6	TC
letrozole	nan	TT
risedronate	LRP5	CC
warfarin	EPHX1	AG
methadone	DRD2	GG
methotrexate	MTHFR	GT

menotropins	FSHR	CC
methadone	KCNJ6	CC
opioids	KCNJ6	CC
methotrexate	MTHFR	GT
ciprofloxacin	CYP1A2	CA
warfarin	CYP4F2	CC

Other

Drug	Gene	Genotype
Dihydropyridine derivatives	NUMA1	GG
etoposide	nan	AA
Dihydropyridine derivatives	CYP3A5	CC
Enzyme inhibitors	CYP19A1	CT
methotrexate	IL12B	TT
methotrexate	FOXP3	GG
atorvastatin, rosuvastatin, simvastatin	APOA5	GA
risperidone	DRD2	TC
methacholine	RUNX1	GA
methacholine	RUNX1	GA
Drugs used in nicotine dependence	CHRNA5	TC

Drugs used in nicotine dependence	CHRNA3, CHRNA5	GA
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Efficacy, Toxicity

Drug	Gene	Genotype
tamoxifen	ABCC2	AA
gemcitabine	SLC28A3	GG
gemcitabine	SLC28A1	CC
gemcitabine	SLC28A2	CA
gemcitabine	SLC28A3	TT
gemcitabine	SLC29A1	GG
gemcitabine	SLC29A1	CC
aripiprazole	FAT1	TT
aripiprazole	FAT1	AT
aripiprazole	FAT1	CC
aripiprazole	FAT1	GC
gemcitabine, paclitaxel	CYP1B1	CC

Efficacy, Metabolism/PK

Drug	Gene	Genotype
Antiemetics And Antinauseants	SLC22A1	GG

Antiemetics And Antinauseants	SLC22A1	CC
irbesartan	APOB	GG
metformin	SLC47A2	GG
tacrolimus	CYP3A5	CC
tacrolimus	CYP3A5	CC
tacrolimus	CYP3A5	CC
montelukast	SLCO2B1	GG
efavirenz	ABCB1	GG
efavirenz	ABCB1	GG
etonogestrel	NR1I2	CC
etonogestrel	PGR	CT

Toxicity

Drug	Gene	Genotype
rivaroxaban	CYP3A5	CC
methotrexate	GGH	GG
methotrexate	ABCC2	CT
methotrexate	SLC19A1	TT
carboplatin, paclitaxel	GSTP1	AA
tenofovir	ABCC4	AC

tenofovir	ABCC4	TT
ethanol	OPRM1	GA
methotrexate	MTHFR	GT
methotrexate	MTHFR	GA
methotrexate	ABCC2	AG
methotrexate	ABCG2	GG

Other, Metabolism/PK

Drug	Gene	Genotype
atazanavir	ABCB1	GG
tacrolimus	CYP3A4	CC
paclitaxel	CYP2C8	CC
paclitaxel	CYP2C8	GG
phenytoin	CYP2C9	GG
ildenafil	CYP3A5	CC
gemcitabine	CDA	GG
phenytoin	CYP2C9	CC
docetaxel	SLCO1B3	AA
docetaxel	SLCO1B3	AA
docetaxel	SLCO1B3	AA

nicotine	UGT2B10	GG
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Dosage, Metabolism/PK

Drug	Gene	Genotype
paclitaxel	CYP2C8	TT
tacrolimus	CYP3A5	CC
phenytoin	CYP2C9	CC
carbamazepine	ABCB1	GG
phenytoin	ABCB1	GG
cyclosporine, tacrolimus	ABCB1	GG
tacrolimus	CYP3A5	CC
lamotrigine	UGT2B7	TC
antipsychotics	ABCB1	GG
warfarin	VKORC1	GA
warfarin	VKORC1	CC
cyclosporine	CYP3A5	CC

Toxicity, Metabolism/PK

Drug	Gene	Genotype
gemcitabine	CDA	AA

imatinib	ABCB1	GG
Arsenic compounds	CBS	AG
Arsenic compounds	CBS	TT
dichloroacetic acid	GSTZ1, POMT2	CT
dichloroacetic acid	GSTZ1, POMT2	GG
dichloroacetic acid	GSTZ1, POMT2	GG
irinotecan	NR1I2	GA
imatinib	SLCO1A2	CT
imatinib	SLCO1B3	GG
isoniazid	NAT2	TT
isoniazid	NAT2	AG

Dosage, Efficacy

Drug	Gene	Genotype
clopidogrel	CYP2C19	CC
ziprasidone	EHF	GG
phenprocoumon	GGCX	TC
warfarin	CYP4F2	CC
warfarin	VKORC1	AG
warfarin	VKORC1	CC

repaglinide	NOS1AP	TG
peginterferon alfa-2a, peginterferon alfa-2b, ribavirin	IFNL3, IFNL4	CC
peginterferon alfa-2a, peginterferon alfa-2b, ribavirin	IFNL3, IFNL4	CC
morphine	COMT	TC
morphine	COMT	AG
morphine	COMT, TXNRD2	TT

Efficacy, Other

Drug	Gene	Genotype
clopidogrel	CYP2C19	CC
clopidogrel	ABCB1	GG
nicotine	CHRNA3	AA
nicotine	CHRNA3	AA

PD

Drug	Gene	Genotype
exenatide, liraglutide	TCF7L2	TC
exenatide	GLP1R	TC
hydromorphone	OPRM1	GA
salbutamol	ADRB2	AA

exemestane	CYP19A1	AG
exemestane	CYP19A1	CA
grazoprevir	SLCO1B1	GA
grazoprevir	SLCO1B1	CC
grazoprevir	SLCO1B1	CC
salbutamol	VEGFA	CC
fentanyl	ADRB2	CC
fentanyl	OPRM1	GA

Dosage, Toxicity

Drug	Gene	Genotype
abemaciclib, palbociclib, ribociclib	ABCB1	GG
acenocoumarol, phenprocoumon	VKORC1	GA
mercaptopurine	NUDT15	CC
sunitinib	ABCB1	GG
sunitinib	NR1I2	TC

Dosage, Efficacy, Toxicity, Metabolism/PK

Drug	Gene	Genotype
olanzapine	CYP3A43	AG

Dosage, Efficacy, Toxicity

Drug	Gene	Genotype
warfarin	CYP2C9	CC

Efficacy, Toxicity, Metabolism/PK

Drug	Gene	Genotype
methotrexate	SLCO1B1	GG
methotrexate	SLCO1B1	CC
methotrexate	SLCO1B1	GA
methotrexate	SLCO1B1	AA
methotrexate	SLCO1B1	AA
methotrexate	SLCO1B1	CC
cyclophosphamide	GSTA1	CC
cyclophosphamide	ABCC1	GG
cyclophosphamide	CYP3A5	CC
cyclophosphamide	ABCC1	CC
cyclophosphamide	GSTP1	AA